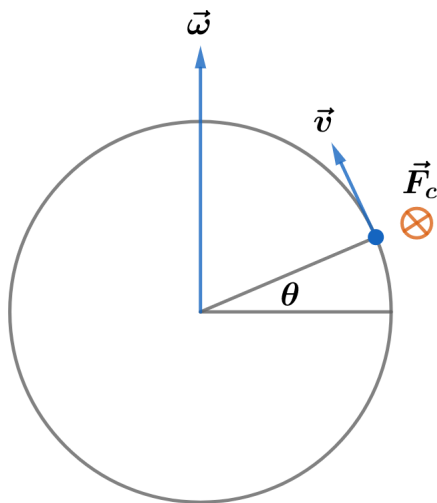


2019A F=ma Exam: Problem 14

Kevin S. Huang



In the rotating frame of the Earth, the Coriolis force is given by

$$\vec{F}_c = 2m\vec{v} \times \vec{\omega}$$

so the bullet gets deflected East. We have

$$a_c = 2v\omega \sin \theta$$

so after time t , the bullet is deflected by

$$\begin{aligned} d &= \frac{1}{2}a_c t^2 = \frac{1}{2}(2v\omega \sin \theta)t^2 \\ &= v \left(\frac{2\pi}{T} \right) t^2 \sin \theta = (200 \text{ m/s}) \left(\frac{2\pi}{1 \text{ day}} \right) \left(\frac{100 \text{ m}}{200 \text{ m/s}} \right)^2 \sin 30^\circ \\ &= 1.8 \text{ mm} \end{aligned}$$

so the answer is D.